

Web Data Integration
Project Outline

**Pharmaceutical Drug Integration: Linking
US, EU and Canadian Approval Data**

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October 6, 2026

Submitted to
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1 Use Case

This project aims to build a unified view of approved pharmaceutical drugs in the United States, the European Union and Canada, enabling researchers and analysts to query regulatory status, approval history, and product attributes across these markets from a single integrated data set. To achieve this, we combine data from three complementary sources. Drugs@FDA is the FDA’s official database of US-approved drugs. EMA Medicines is the European Medicines Agency’s register of EU-authorised medicines. The Health Canada Drug Product Database (DPD) lists all drug products authorised in Canada. The central entity is the active substance, the pharmacologically active ingredient in a drug product. Active substances serve as a natural integration key because they persist across different brand names, manufacturers, regulatory jurisdictions, and formulations. The same substance can even carry different names across regions, as with paracetamol in the EU and acetaminophen in the US and Canada. Beyond regional synonyms, biosimilars carry manufacturer-specific suffixes that further complicate direct matching. Resolving these naming variations through identity resolution is therefore a central task of the project, ultimately producing one clean record per active substance with consolidated attributes from all three sources.

2 Requirements Suitability

2.1 Schema and basic profile

Table 1: Overview of the input datasets

Dataset	Source (*)	Format	# Ent.	# Attr.	List of attributes (**)
Drugs@FDA	U.S. FDA	TXT/TSV	51,864	56	ApplNo, ProductNo, Form, Strength, ReferenceDrug, DrugName, ActiveIngredient, ApplPublicNotes ^{**} , SponsorName, SubmissionType, SubmissionStatus, SubmissionsPublicNotes ^{**} , ReviewPriority ^{**} , MarketingStatusID, TECCode, ...
EMA Medicines	EMA	XLSX	2,746	39	Name of medicine, EMA product number, Medicine status, Opinion status ^{**} , Active substance, Species (vet.) ^{**} , ATC code, Biosimilar, Generic, Orphan medicine, Opinion adopted date ^{**} , Revision number ^{**} , ...
DPD	Health Canada	TXT/CSV	42,042	87	DIN, brand name, ingredient, strength, dosage unit ^{**} , company name, class, ATC code, schedule, status, status date, form, route, descriptor ^{**} , ...

(*) **Source:** Downloaded directly from the respective official provider websites.

(**) **Missingness:** ^{**} indicates >30% missing values. High missingness is mostly structural (e.g., optional notes, category-specific fields, or particular regulatory events).

Table 1 summarises the datasets and their most relevant attributes, including those with high missingness.

Drugs@FDA

Drugs@FDA contains 51,864 product records across 29,377 applications. The product table contains only 8,388 distinct drug names and 2,878 distinct active-ingredients, showing substantial reuse across products. The dataset is strongly imbalanced. Among applications, 78.3% are ANDAs, 20.1% NDAs, and 1.7% BLAs. Three attributes exceed 30% missingness: `AppPublicNotes` (99.99%), `SubmissionsPublicNotes` (99.83%), and `ReviewPriority` (41.73%).

EMA Medicines

The EMA dataset contains 2,746 medicine records and 1,870 distinct active ingredient, with no duplicate rows. It is predominantly composed of human medicines (85.6%). Medicine status is also imbalanced: 68.5% are currently authorised, 15.4% withdrawn, and 10.4% represent withdrawn applications. Fifty medicine names occur more than once, again showing that names alone do not uniquely identify entities. Twelve EMA attributes exceed 30% missingness, mostly due to structural reasons (e.g., veterinary attributes being inapplicable to human medicines).

Health Canada Drug Product Database (DPD)

The Health Canada Drug Product Database lists all drug products authorised in Canada in 12 tables linked by `DRUG_CODE`. Its 58,137 products (DINs) form 42,042 medicines (brand name + active substances) with 5,763 distinct active ingredients. 87.7% are human products; 66.5% are cancelled and 23.0% marketed. Ingredients follow a “base (salt)” pattern, e.g. `ESOMEPRazole` (`ESOMEPRazole` `MAGNESIUM TRIHYDRATE`). Notable missingness: ATC code (21.47%) and first marketed date (11.67%).

2.2 Integrated schema and overlap with input schemata

Because the entity is the active ingredient, the integrated schema aggregates product-level information from all three sources into one record per ingredient. Required transformations include splitting the Drugs@FDA Form field, separating the DPD “base (salt)” ingredient names, and breaking down combination products. The integrated schema comprises 15 attributes (Table 2).

Table 2: Attribute Intersection with Integrated Schema

Attribute name	Attribute type	Datasets in which the attribute is found
activeIngredientName	string	Drugs@FDA, EMA Medicines, DPD
brandNames	list (string)	Drugs@FDA, EMA Medicines, DPD
manufacturers	list (string)	Drugs@FDA, EMA Medicines, DPD
marketingStatus	string (categorical)	Drugs@FDA, EMA Medicines, DPD
productClass	string (categorical)	Drugs@FDA (derived), EMA Medicines, DPD
atcCode	string	EMA Medicines, DPD
biosimilar	boolean	EMA Medicines, DPD
generic	boolean	Drugs@FDA (derived), EMA Medicines
dosageForms	list (string)	Drugs@FDA, DPD
routes	list (string)	Drugs@FDA, DPD
strengths	list (string)	Drugs@FDA, DPD
usApprovalDate	date	Drugs@FDA
euAuthorisationDate	date	EMA Medicines
caApprovalDate	date	DPD

activeIngredientName is the primary matching key. **brandNames** and **manufacturers** occur in all three datasets and are modelled as lists (fusion by voting). **marketingStatus** assigns each ingredient to one of five classes, derived from source-specific status flags (fused by precedence: Active > Suspended > Pending > Discontinued > Not authorised). **usApprovalDate**, **euAuthorisationDate** and **caApprovalDate** describe distinct regulatory events and stay separate. **productClass** is Human for all Drugs@FDA records and serves as a blocking key; **generic** is derived from the Drugs@FDA application type (ANDA). **atcCode** is language-independent and helps match synonyms such as paracetamol and acetaminophen (both N02BE01).

2.3 Entity Count Requirement

The three datasets cover overlapping parts of the pharmaceutical market: each registers the medicines approved by its own regulator, so many medicines are marketed in two or all three regions. Modelling the medicine as brand name + active substance(s), we build a key **medicine_id** per medicine (upper case, strengths removed from the brand name, salt words removed from the substances, e.g. NEXIUM|ESOMEPRAZOLE). Matching this key exactly, 1,880 medicines appear in at least two datasets and 408 in all three. Pairwise, Drugs@FDA and the DPD share 1,470 medicines, EMA and the DPD 701, and Drugs@FDA and EMA 525.

Exact matching still misses harder cases. Drugs@FDA lists NEXIUM with ESOMEPRAZOLE MAGNESIUM, the DPD lists NEXIUM - 20MG with ESOMEPRAZOLE (ESOMEPRAZOLE MAGNESIUM TRIHYDRATE), and EMA lists the over-the-counter Nexium Control (esomeprazole) from a different licence holder.